

Project Submission

Exploring the Power of Agentic AI with IBM Granite and watsonx Orchestrate

Project Title: Interview Trainer Agent | **Problem Statement No. 22**

Domain: Education & Career Development (HR Technology)

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Problem Statement No. 22 — Interview Trainer Agent



The Challenge

An Interview Trainer Agent, powered by RAG (Retrieval-Augmented Generation), prepares users for job interviews by generating tailored question sets and preparation strategies based on their profile name, experience level, and job role.

It retrieves role-specific interview questions, industry expectations, behavioral scenarios, and HR guidelines from recruitment portals, professional networks, and company interview databases.

Users can input their resume or job title, and the agent provides targeted questions, model answers, and improvement tips.

It supports both technical and soft-skill assessment, ensuring a comprehensive interview prep experience.

This AI-driven assistant builds user confidence, sharpens responses, and increases success rates in competitive hiring environments.

Proposed Solution



The Interview Trainer Agent is built on IBM watsonx Orchestrate as a RAG-powered conversational agent that delivers personalized, role-specific interview coaching in real time.



Resume-Grounded Knowledge

The candidate's resume is uploaded as a knowledge source, so every question set is grounded in their real projects, skills, and academic background.



Conversational Interface

A chat-based agent greets the user with quick-start prompts — mock interviews, prep strategies, and behavioral questions — for a guided experience.



ReAct Reasoning Style

Configured with the ReAct agent style so the model can plan, retrieve, and refine its response before delivering tailored guidance.



Multi-Channel Delivery

Deployable across Teams, WhatsApp, Slack, Facebook Messenger, SMS, and voice/phone channels for accessible, anywhere practice.

Technology Used



IBM watsonx Orchestrate



Low-code platform used to build, configure, and deploy the Interview Trainer Agent (built and hosted on IBM Cloud).

IBM Cloud Lite Services



Provides the underlying trial infrastructure to build, test, and run the agent in a live environment.

Foundation Model — GPT-OSS 120B



Powers the agent's reasoning and personalization; watsonx Orchestrate also supports IBM Granite models for the same workflow.

RAG (Retrieval-Augmented Generation)



Grounds every response in the candidate's uploaded resume for accurate, personalized question generation.

Agent Behavior & Guidelines Engine



Custom instructions define the agent's persona, experience-level context, and interview-prep methodology.

Multi-Channel Connectors



Teams, WhatsApp (Twilio), Slack, Facebook Messenger, SMS, and phone/voice channels for flexible access.

watsonx Orchestrate Components Used



Profile

Agent description, welcome message, and quick-start prompts (mock interview, prep strategy, behavioral questions).

Knowledge

Resume uploaded as a knowledge source ("Resume — by IBMid") so answers stay grounded in the candidate's real background.

Behavior

Custom instructions define the agent's role as an Interview Trainer Agent and its RAG-based prep strategy; Agent Style set to ReAct.

Toolset

Extendable with organizational tools so the agent can take further action beyond conversation.

Channels

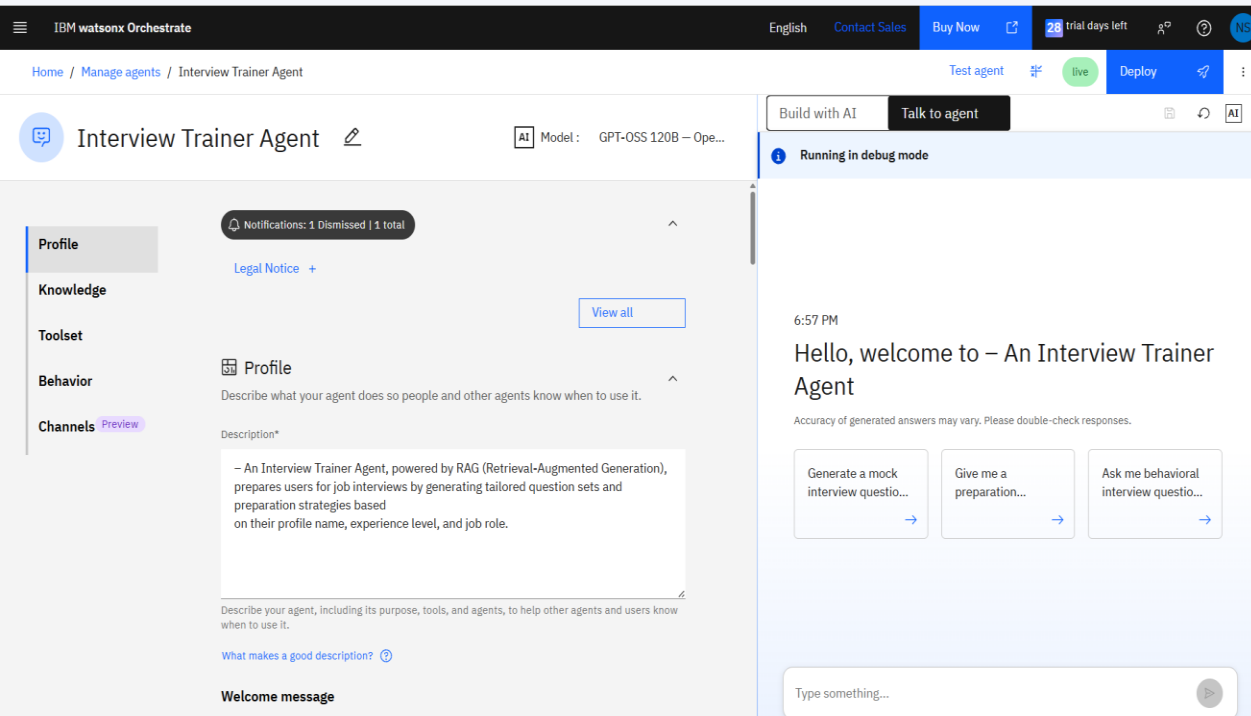
Home page, Embedded agent, Teams, WhatsApp with Twilio, Facebook Messenger, Slack, Phone (Genesys/SIP), and SMS.

Deployment

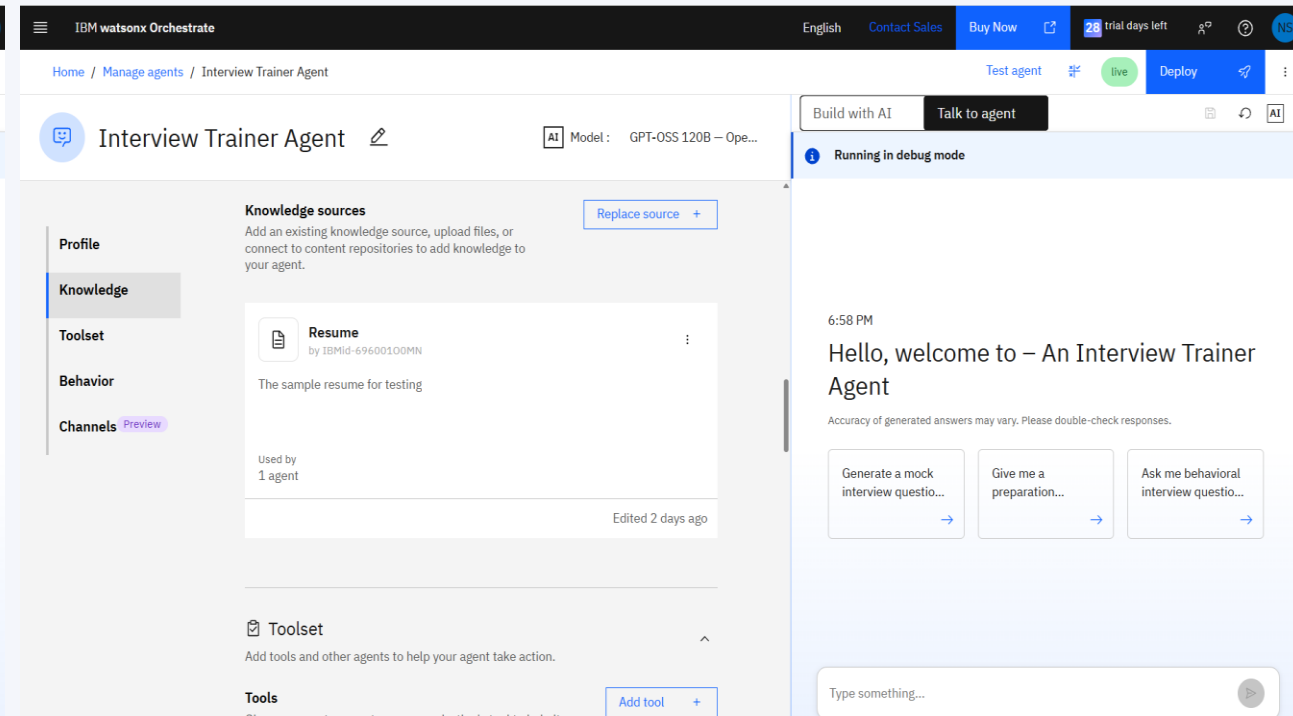
Agent tested in Debug mode via "Talk to agent" and deployed live from Manage Agents.

Agent Build — Profile & Knowledge Source

IBM watsonx Orchestrate: Interview Trainer Agent configuration



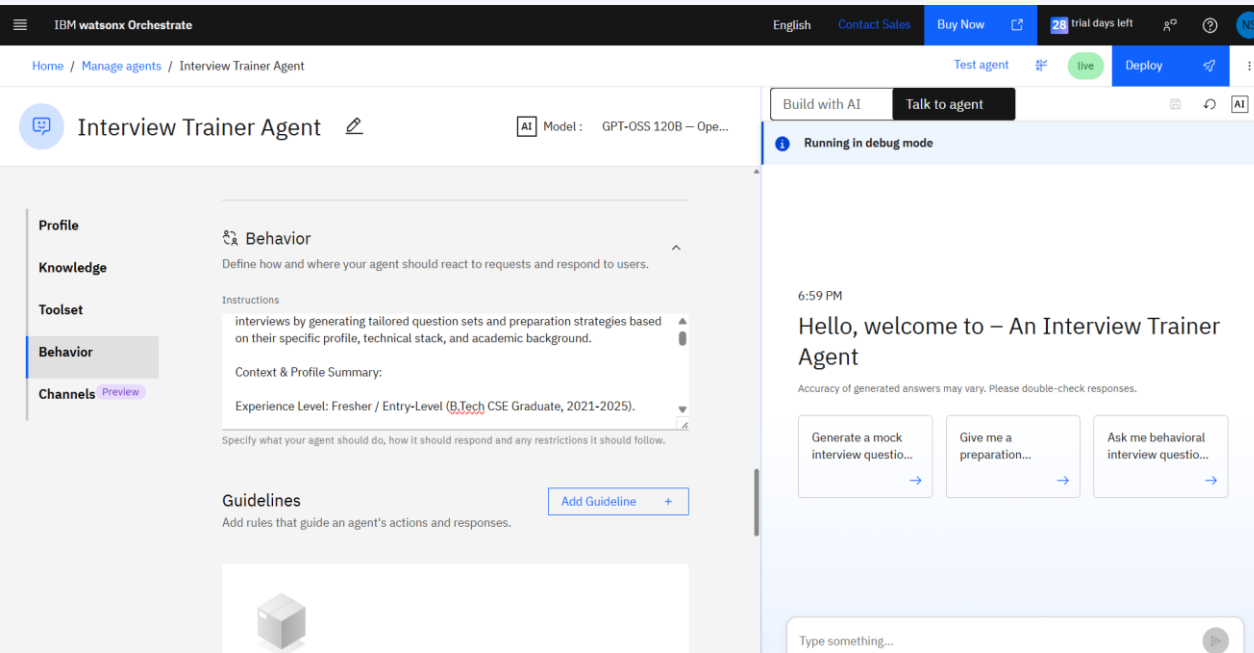
Left: Agent profile describing purpose — generating tailored interview question sets based on name, experience level, and job role.



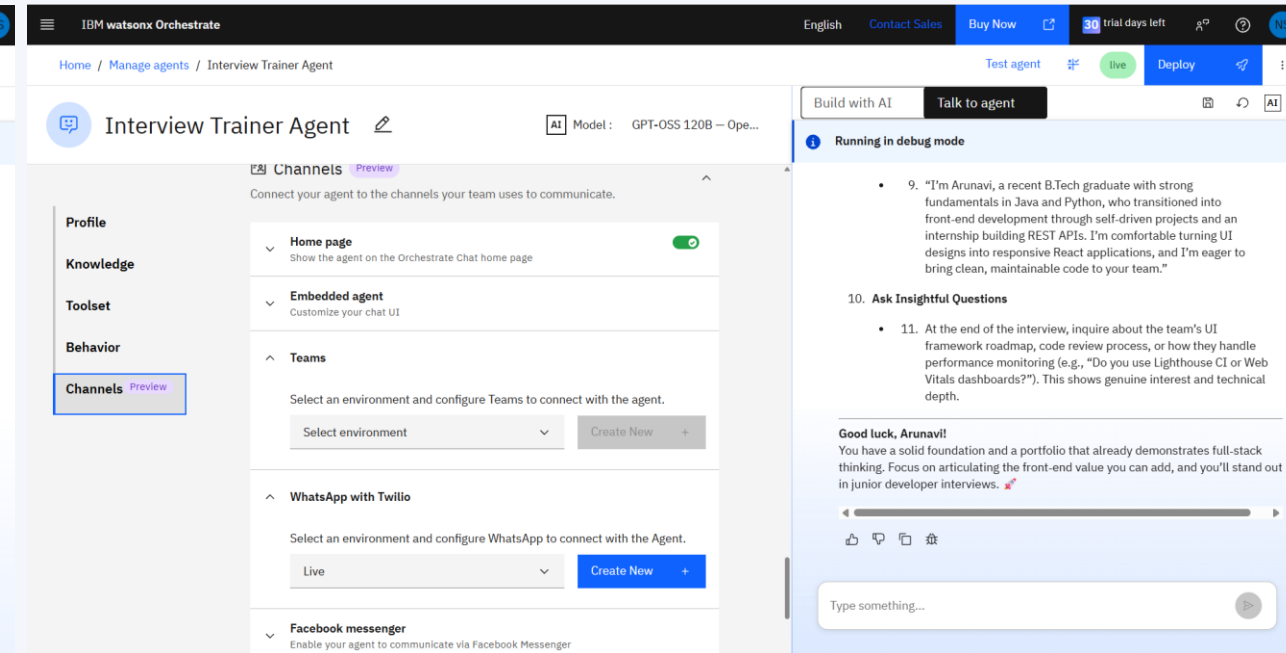
Right: Candidate resume added as a knowledge source, enabling RAG-based, resume-grounded responses.

Agent Build — Behavior & Multi-Channel Deployment

IBM watsonx Orchestrate: Interview Trainer Agent configuration



Left: Behavior instructions defining the agent's context, profile summary, and interview-prep strategy.



Right: Channels enabled for Teams and WhatsApp with Twilio, alongside Facebook Messenger, Slack, phone, and SMS.

Live Demo — Personalized Interview Prep Output

Test conversation run in watsonx Orchestrate debug mode

Running in debug mode

You 6:59 PM

Generate a mock interview question set focused on my Java & MySQL Online Food Ordering project.

Interview Trainer Agent 6:59 PM

Custom Interview Prep Plan: Software Engineer (Fresher)

Candidate Profile: Arunavi

1. Target Tech Stack Focus Strategy

Area	What to Review	How to Practice
Core Java	• OOP fundamentals (encapsulation, inheritance, polymorphism, abstraction) • Collections Framework (List, Set, Map, Queue) • Exception handling,	• Re-implement a simple CRUD service using plain JDBC • Solve 5–7 medium-level LeetCode problems that require using <code>HashMap</code>,

Save as test

Type something...

Build with AI

Talk to agent

Running in debug mode

2. Customized Question Set

Data Structures & Algorithms (DSA)

1. Problem: "Given a list of orders, each with a timestamp and total amount, find the maximum total amount of orders placed within any sliding window of 30 minutes."

- 2. Why it fits: Tests your ability to use a **Deque/Sliding Window** technique, which is common in order-processing systems (e.g., peak-load detection).
- 3. Follow-up: Explain time/space complexity and how you would adapt the solution if the data were streamed from a database.

4. Problem: "Design a data structure to efficiently retrieve the top-k most-ordered dishes in real time."

- 5. Why it fits: Requires a **Heap (PriorityQueue)** or **TreeMap** with frequency counting—directly relevant to a food-ordering platform's analytics.

Core Technical (Java / SQL / DBMS)

Save as test

Type something...

Running in debug mode

Project & Internship Drill-down (Online Food Ordering System)

1. Scenario Question: "During your internship you built REST APIs for order management. Walk me through the end-to-end flow of placing an order in your Online Food Ordering project, covering:
a) API endpoint design (HTTP method, URI)
b) Input validation and error handling
c) Transaction management with MySQL
d) Response structure (status codes, payload). Also, describe a bug you encountered in this flow and how you resolved it."

- 2. What it tests: Understanding of REST principles, Java-Spring (or servlet) implementation, JDBC transaction control, and problem-solving mindset.

Behavioral (STAR Method)

1. Question: "Tell me about a time you participated in a hackathon (or coding competition) where you had to deliver a functional prototype under a tight deadline. What was your role, what challenges did you face, how did you organize the work, and what was the outcome?"

- 2. Goal: Evaluate teamwork, time management, ability to work under pressure, and learning orientation. Use the **STAR** framework (Situation, Task, Action, Result) in your answer.

Save as test

Type something...

Prompt: "Generate a mock interview question set focused on my Java & MySQL Online Food Ordering project." The agent returns a candidate-specific prep plan — tech-stack focus areas, a customized DSA/technical question set, a project & internship drill-down, and STAR-method behavioral questions.

Role of Agentic AI in the Solution

- Acts as an autonomous interview coach rather than a static Q&A chatbot — planning, retrieving, and reasoning before responding.
- Uses the ReAct agent style to think, act, observe, and refine its approach until a complete, tailored prep plan is delivered.
- Grounds every recommendation in the candidate's resume via RAG, bringing genuine context awareness to each session.
- Adapts across technical, project-based, and behavioral (STAR) question types within a single continuous conversation.
- Is reachable across multiple channels — chat, Teams, WhatsApp, Slack, SMS, and voice — making practice available anywhere.

Novelty and Uniqueness



Resume-Grounded RAG

Question sets are generated from the candidate's actual resume, not generic templates — every question maps to a real project or skill.



Full-Spectrum Coverage

A single agent spans technical, project-drill-down, and behavioral (STAR) preparation in one conversation.



Low-Code, Enterprise-Ready

Built entirely on IBM watsonx Orchestrate — no custom backend required, with governance and scalability built in.



Omni-Channel Access

Deployable to Teams, WhatsApp, Slack, Messenger, SMS, and voice, meeting candidates on the channel they already use.

GitHub Repository



Repository format: public GitHub repository with README enabled.

<https://github.com/naveenasandhi/Interview-Trainer-Agent/tree/main>

Repository must include:

- app.json — exported watsonx Orchestrate agent configuration
- Interview_Trainer_Agent_Problem_Statement.pdf — problem statement document
- Interview_Trainer_Agent_Presentation.pptx — this project presentation
- Any additional supporting files (screenshots, sample resume, README)

Demo-Video-Link

<https://drive.google.com/file/d/1IPOMpoFME3roxfRoOfmgHgAeUUIf7wUL/view?usp=drivesdk>

This video demonstrates the Interview Trainer Agent built using IBM watsonx Orchestrate.

Future Scope



Voice-Enabled Mock Interviews

- 1 Activate watsonx Orchestrate's voice modality so candidates can rehearse spoken answers and receive tone/pacing feedback, closer to a real interview panel.

Recruiter & Job-Description Matching

- 2 Let the agent ingest a target job description alongside the resume to highlight gaps and tailor question sets to that specific role.

Performance Analytics Dashboard

- 3 Track a candidate's practice history, frequently missed question types, and improvement over multiple sessions.

Deeper Toolset Integration

- 4 Connect calendar and ATS tools so the agent can schedule mock sessions and pull live job postings automatically.

Certificates Earned

Credly Certificate

In recognition of the commitment to achieve professional excellence



Naveena Sandhi

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



Issued on: Jun 22, 2026

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IBM BOB Certificate

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Naveena Sandhi

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 22 Jun 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.